

GxP Training Bot

AI-Assisted Adaptive Training and Assessment Platform for Pharmaceutical SOPs

Student: _____ Roll No.: _____ Guide: _____ Period: 13 Jul – 14 Aug 2026

PROJECT IN ONE PARAGRAPH

The platform converts controlled pharmaceutical SOPs into SME-approved assessments and uses each learner's section-level performance to determine subsequent training content. A large language model drafts questions from one SOP section at a time; every draft requires SME approval under an electronic signature bound to a hash of the approved content. Because each question retains a reference to its source section, every response is evidence about a specific passage, so retraining targets measured weakness instead of repeating the whole procedure.

VERIFIED STATUS

Verification	Result	Verification	Result
Backend test suite	221 passing, 0 failures, 0 skipped	Frontend static analysis	0 errors, 0 warnings
Migration check	No outstanding changes	Frontend build	Successful
Deployment check	No issues identified	End-to-end workflow	Verified on live NVIDIA NIM
Functional requirements	80 of 84 implemented (95.2%)	Development period	11 commits, 13 Jul – 14 Aug 2026

VERIFIED CONTROLLED DEMONSTRATION RESULT

Measure	Value
Questions generated (3 sections × 3)	9
First-attempt score	33.33% (3 of 9)
Questions selected for retraining	6 of 9 — the section already mastered was excluded
Learning gain, both weak sections	50% → 100% (+50 percentage points)

After retraining the recency-weighted score was 87.9% against a lifetime accuracy of 75.0%, the weighting registering the improvement. This is a controlled demonstration result, not a statistical user study; no claim of population-level effectiveness or statistical significance is made.

ADAPTIVE ENGINE — CLASSIFICATION

Layer	Mechanism	Learns from learner data?
1 — Adaptive priority	Rule-based: thresholds 60/80%, minimum evidence 3, mastery streak 3	No — values chosen, not fitted
2 — Performance signal	Statistical: recency-weighted accuracy, half-life 5 responses	Yes — computed from responses
3 — Elo	Online estimation of learner ability and item difficulty ($K = 32 / 16$)	Yes
4 — FSRs	DSR memory model governing review timing	Yes (state); weights are fixed defaults

Hybrid adaptive system. The large language model does not determine adaptive priority; the adaptive engine is not a trained neural network. The system is adaptive because measured performance changes future content selection.

PROGRESS BY PHASE

Phase	Milestone	Phase	Milestone
1 (13 Jul)	Requirements, architecture, core platform	6 (4 Aug)	Section-level mastery introduced
2–3 (21 Jul)	SME workflow, e-signature, first adaptive retraining	7 (9–14 Aug)	Requirements audit; 84 requirements traced
4 (21 Jul)	Retrieval-augmented SOP chatbot	8 (14 Aug)	Recency weighting, evidence gate, hardening
5 (26 Jul)	Elo estimation; FSRs replaces fixed intervals	9–10 (14 Aug)	Adversarial verification; end-to-end validation

KEY DEFECTS FOUND AND CORRECTED

- Correctly answered sections were still retrained, and never-assessed sections were invisible to selection — selection reimplemented on observed accuracy, with never-assessed made an explicit HIGH state.
- Whole-document mastery concealed weak sections — resolved by introducing section-level mastery.
- The answer key was returned to learners, a completed attempt could be resubmitted, and approved content could be modified through the API — role-selected serialisers, an atomic claim returning HTTP 409, and HTTP 403 with content-hash verification.
- Duplicate question identifiers fabricated evidence — rejected with HTTP 400 before any database write.

REMAINING WORK

Priority	Item	Effect
1	Persist the offered question set per attempt	Makes the adaptive decision enforced rather than advisory; attempts reproducible
2	SOP version lifecycle	Removes the only gap that can silently destroy learner mastery data
3	Question revision history	Completes content lineage; signature binds to a version
4	Difficulty-aware adaptive priority	Removes an inconsistency in how responses are weighted
5	Entailment verification of generated items	Upgrades grounding from provenance to verification

STATUS

All seven stated objectives are implemented and verified. The complete workflow has been verified end to end against the live model provider, and 80 of the 84 specified functional requirements are implemented; the four outstanding items are documented as future scope.